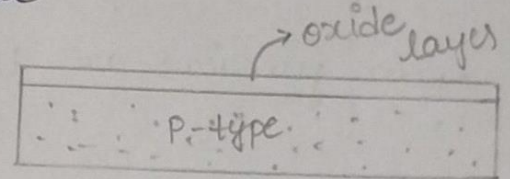


- 1) Explain the process flow of NMOS transistor - p-type substrate with equivalent diagrams.

PROCEDURE

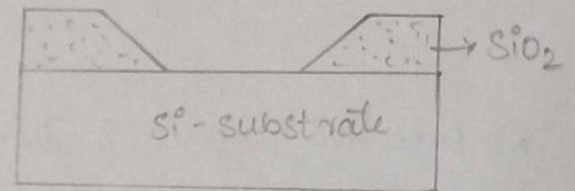
- 1) Silicon Substrate Oxidation - The process begins with a p-type silicon substrate. This substrate is then oxidized, means a layer of SiO_2 is grown on the surface.

This thick oxide is called field oxide (1 μm)



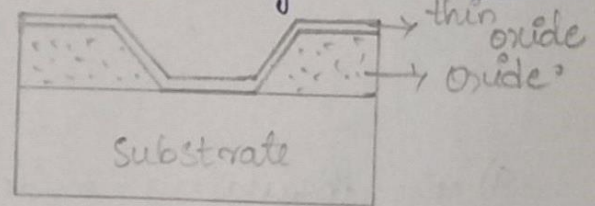
- 2) Etching

This etching process removes the oxide from the areas where the MOS transistor will be built, exposing the bare silicon substrate in those regions.



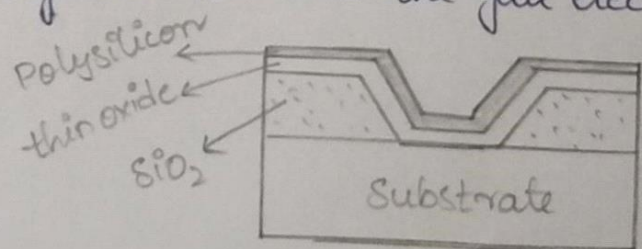
- 3) Gate oxide Growth

A thin, high quality layer of silicon dioxide is grown on the exposed silicon surface. This thin oxide layer will become the gate oxide of MOS.



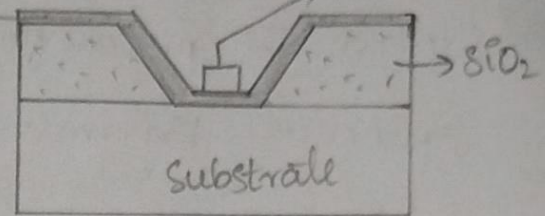
- 4) Polysilicon Deposition

A layer of polysilicon is deposited over the entire surface, covering the thin gate oxide. Polysilicon serves as the gate electrode material for MOS.



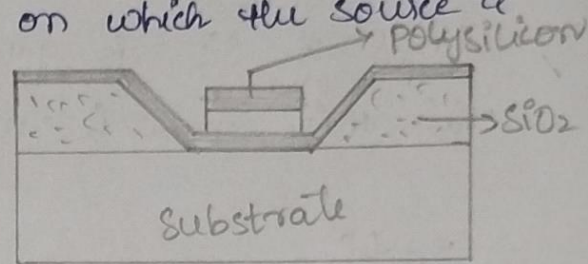
5) Polysilicon Patterning & Etching

After deposition, the polysilicon layer is patterned and etched to form gate of the mos. thin oxide

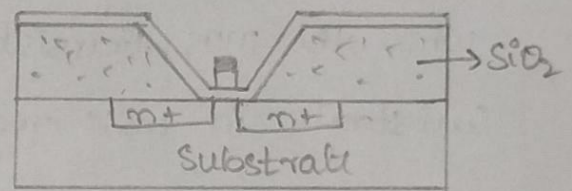


6) Gate oxide Etching.

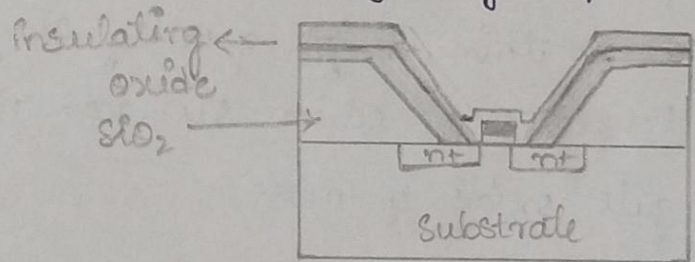
The gate (thin) oxide not covered by polysilicon is also etched away, which exposes the silicon surface on which the source & drain junctions are to be formed.



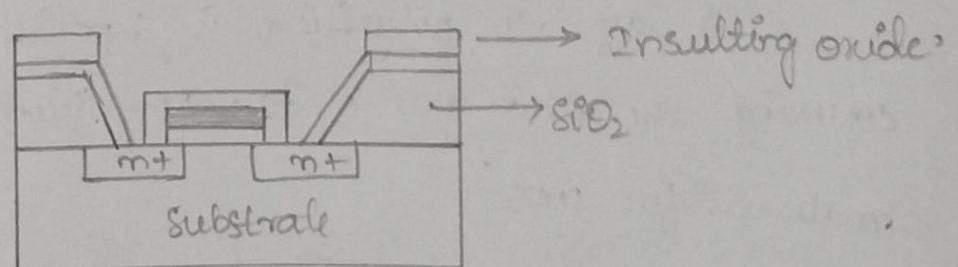
7) The entire silicon surface is then doped with a high conc of impurities ultimately creating two



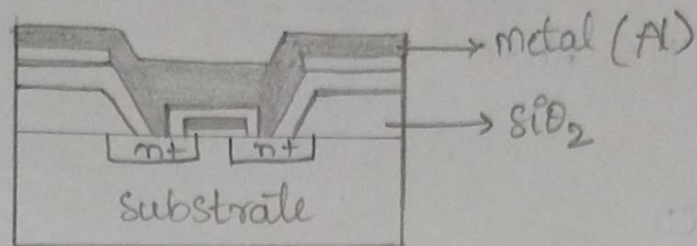
8) Once the source & drain regions are completed, the entire surface is again covered with an insulating layers of silicon dioxide.



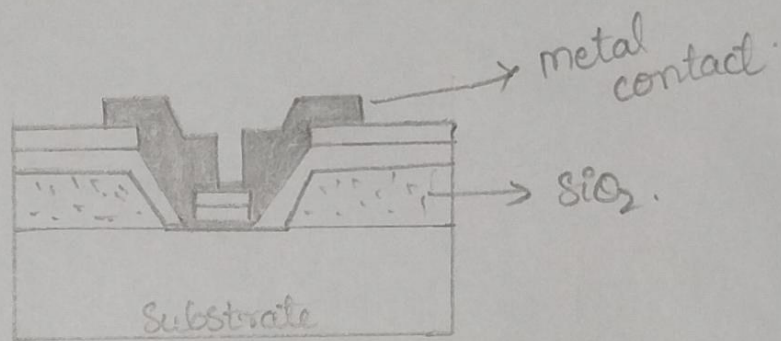
9) The insulating oxide layer is then patterned in order to provide contact windows for the drain & source.



10) then the surface is covered with evaporated aluminum which will form the interconnects.



11) Finally, the metal layer is patterned & etched, completing the interconnection of the MOS transistor on the surface.



MOS transistors, usually a second & third layer of metallic interconnect can also be added on top of this structure by creating another insulating oxide layer, cutting contact via holes, depositing & patterning the metal.

2.) Draw the layout for the following boolean expression,

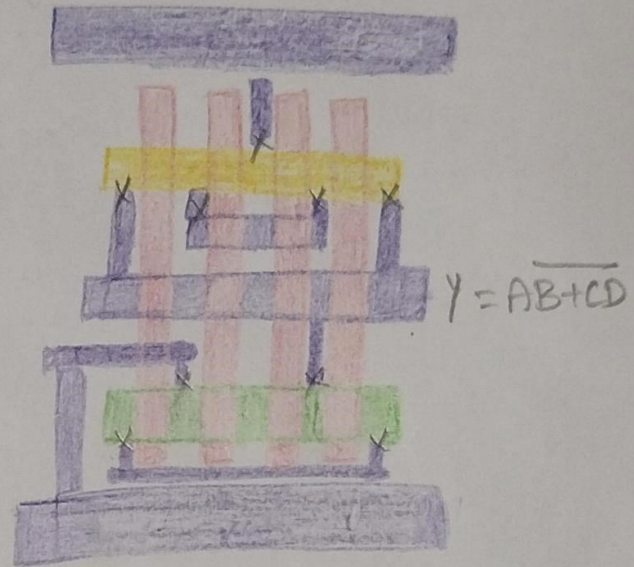
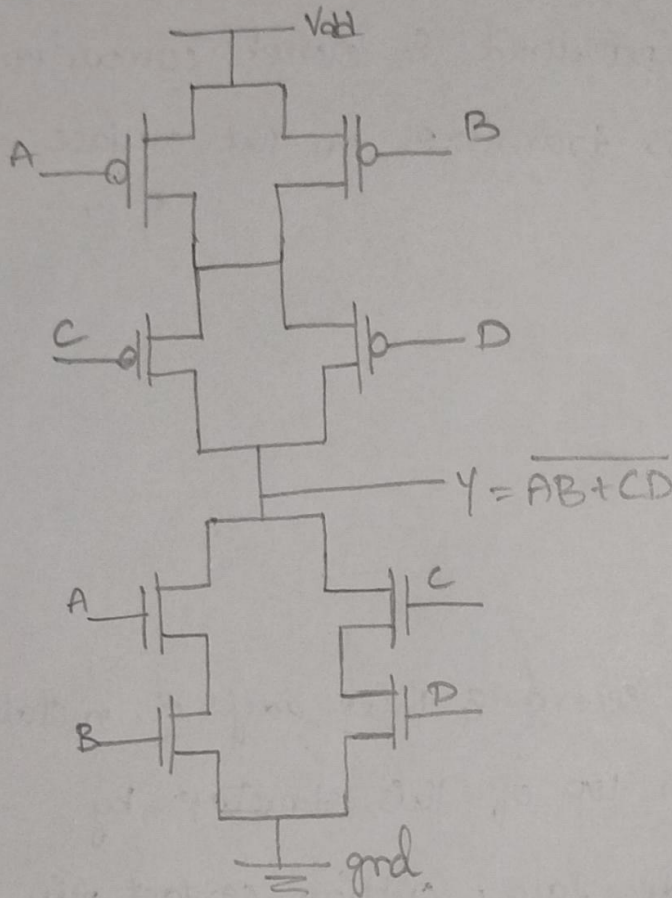
1) $Y = \overline{AB+CD}$

2) $Y = \overline{(A+B)CD}$

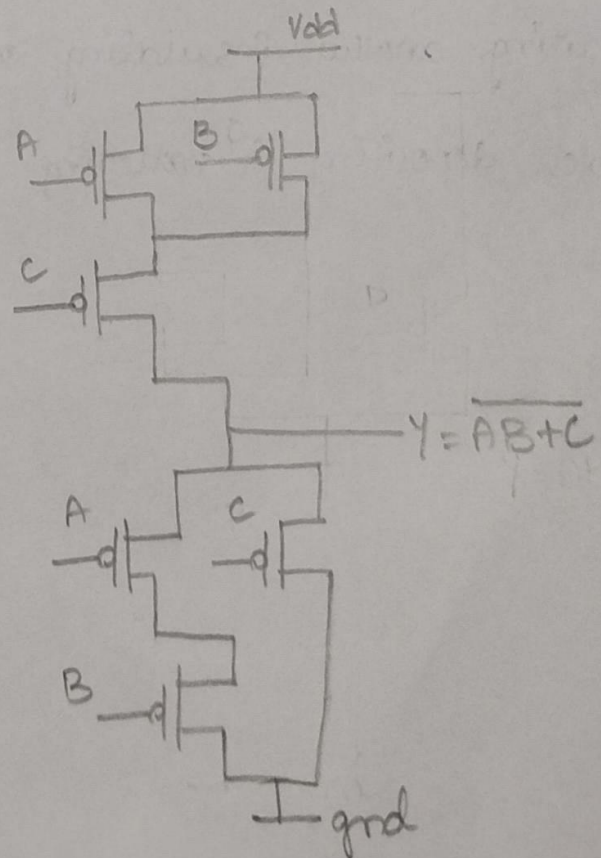
3) $Y = \overline{AB+C}$

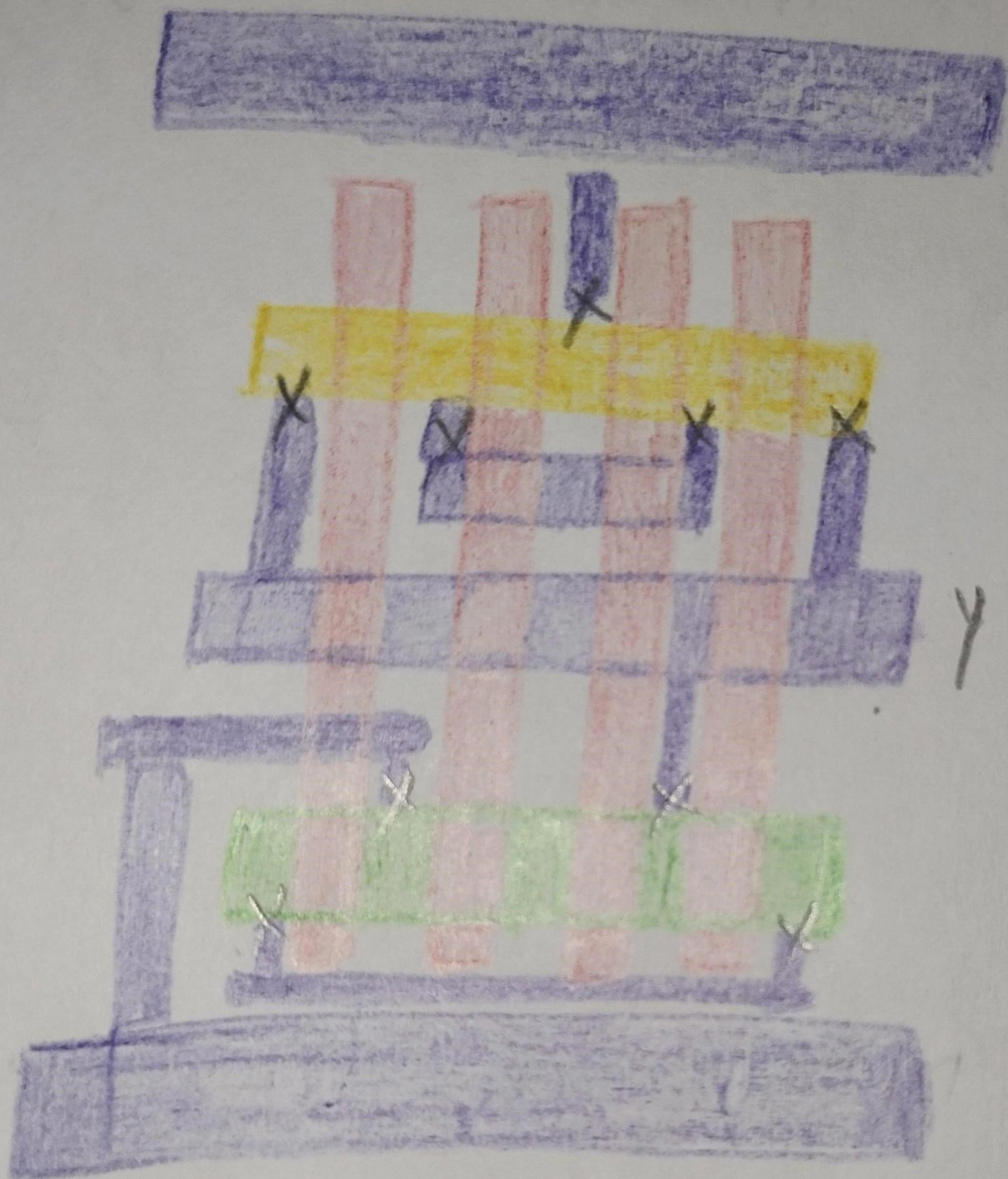
- PMOS - parallel
- NMOS - series
- + NMOS - parallel
- PMOS - series

① $Y = \overline{AB+CD}$



③ $Y = \overline{AB+C}$





$$Y = \overline{A}B + CD$$

$$\textcircled{2} \quad Y = \overline{(A+B)CD}$$

30) List the steps involved in patterning of silicon dioxide type Insulator substrate.

① Photoresist Application

The Silicon wafer with silicon dioxide layer is coated with a thin layer of photoresist. where photoresist is a light sensitive material that changes its solubility when exposed to light.

② Mask Alignment & Exposure

A photomask, which contains the desired pattern, is placed over the wafer. UV light is shone through the mask, exposing the photoresist in the areas not blocked by the mask.

③ Photoresist Development

The wafer is immersed in a developer solution. Depending on the type of photoresist used the (+ve & -ve) either the exposed or unexposed portions of the photoresist are dissolved, leaving behind the desired pattern on the wafer.

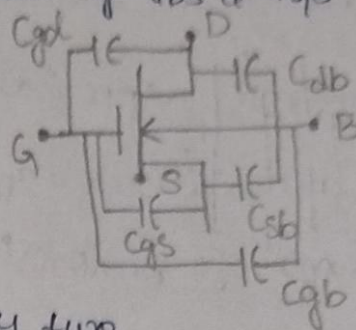
④ Oxide Etching - wet, Dry

In wet Etching the wafer is immersed in a chemical solution, that selectively removes the silicon dioxide in the areas not protected by the remaining photoresist. In Dry Etching the wafer is placed in a plasma chamber where reactive gases are used to etch away the silicon dioxide.

⑤ Photoresist Stripping

After the oxide etching is complete, the remaining photoresist is removed from the wafer using a stripping solution. This leaves behind the patterned silicon dioxide features on the silicon substrate.

4) Identify the Capacitances in MOSFET and explain how it's going to effect the characteristics of MOS with respect of V_{DS} & V_{GS} .



There are 5 capacitors in MOSFET i.e.,

C_{gd} , C_{db} , C_{gs} , C_{sb} , C_{gb} .

Capacitor - passive element separated, by two

conductors separated by an insulating material called as dielectric.

Gate-Source Capacitance (C_{gs})

Cutoff region: $V_{GS} < V_{th}$ - MOSFET is off so essentially C_{gs} is very small or zero.

Triode: $V_{GS} > V_{th}$, $V_{DS} < V_{GS} - V_{th}$ - A channel forms, C_{gs} increases.

Saturation: $V_{GS} > V_{th}$, $V_{DS} > V_{GS} - V_{th}$ - Channel is pinched off near the drain. C_{gs} is roughly $2/3$ of gate channel capacitance.

Gate-Drain Capacitance (C_{gd})

Cutoff region: C_{gd} is close to zero.

Triode: C_{gd} significantly ~~increases~~ increases, ideally becoming ^{not} zero.

Saturation: C_{gd} significantly decreases, ideally becoming zero.

V_{DS} - C_{gd} is strongly influenced, as V_{DS} increases the channel pinch off becomes more pronounced, reducing C_{gd} .

Drain-Body Capacitance (C_{db})

V_{DS} dependence - C_{db} is a junction capacitance associated with the reverse biased. As V_{DS} increases, the reverse bias across the bd increases leads to C_{db} decreases.

Source - Body Capacitance (C_{sb})

C_{sb} is influenced by the potential difference b/w the S & b. In typical operation, the source is connected to ground so, V_{sb} is constant.

Gate - Body Capacitance (C_{gb})

Cut off : C_{gb} is the dominant gate capacitance

Triode : C_{gb} decreases.

Relation C & V in MOSFET

Parameter	Cutoff	Triode	Saturation
C_{gb}	C_0	0	0
C_{gd}	0	$C_0/2$	0
C_{gs}	0	$C_0/2$	$3C_0/2$
C_{gin}	C_0	C_0	$3C_0/2$

5) Discuss about accumulation, depletion & Inversion mode of MOSFET.

Accumulation mode

This occurs when a voltage is applied to the gate that attracts majority carriers towards the gate. For n-channel, this means applying a negative voltage to gate. This negative voltage attracts holes from p-type towards the gate, accumulating them at the surface. This creates a region with a higher conc of holes near the gate, but a conducting channel is not created b/w source & drain.

① No channel is formed, very little current b/w S & D.

② Gate with majority holes

Depletion mode

This mode happens when a gate voltage is applied that repels majority carriers away from the gate region, creating a depletion region. Positive gate voltage repels holes from p-type body, creating a region near the gate that's depleted of mobile charge carriers. This depletion region acts as an insulator, further preventing current b/w S & D.

① Depletion region near gate

Inversion mode

This mode is achieved when the gate voltage is increased further in the same polarity as the depletion mode, causing the region near the gate to become 'inverted' with minority carriers. As the positive gate voltage increases, it attracts e^- from source & drain towards the gate. When V_{gate} exceeds V_{th} enough e^- accumulate near the gate to form a continuous conducting channel b/w source & drain.

mode	Gate voltage	Carrier	channel formation	Current
Accumulation	Negative	majority charge carriers	No	OFF
Depletion	Small Positive	majority charge repelled, depletion region forms	No	OFF
Inversion	sufficiently positive	channel forms.	Yes	ON

Draw the layout of a two input NAND gate using λ based design rules.

